

Time Integrity from a Shared Geophysical Signal: Cross-Node Correlation and Sequential Detection of Delay Attacks

Kyle Luecke

Abstract—Two-way time transfer is rank deficient: one exchange measures the sum of the forward and reverse path delays, but only their difference enters the offset estimate, so an adversary delaying one direction by δ shifts the estimated offset by $\delta/2$ while every packet stays cryptographically authentic. We close the gap with a globally excited, spatially correlated geophysical signal, the 6 to 25 Hz Schumann band, recorded independently at each node and exchanged inside authenticated, acquisition-timestamped records, so that the timing fiducial rides in message content and out of reach of transport delay. Fitted mode parameters on a four-year record give a 90 s integrity floor; the raw waveform supports 12 to 363 μ s at assumed unity coherence. We then replace assumption with measurement. Public magnetometer records from observatories 3002 km apart show fundamental-mode coherence that survives a pre-declared null battery and disjoint-epoch replication, bounding offset precision at 184 μ s per day through a facility sensor noisier than the signal. A matched-sensor test, frozen and hash-deposited before the data were seen, confirmed its preregistered prediction, with coherence rising fifty-fold inside a forecast window of thirty to one hundred, for a measured 76.6 μ s bound at six hours and a projected 38 μ s at one day. A separately preregistered lag-stability criterion was not met: the lag is detected in nearly every window but wanders with source geometry, a measured limitation retained for the named follow-on rather than relaxed. Against slow attacks, sequential detection limits pre-alarm bias to $\theta_{\text{det}} = (\kappa\sigma\sqrt{3\Delta t})^{2/3}r^{1/3}$, so patience buys the adversary less, not more. Every empirical claim in this paper is bound to a public preregistration hash.

Index Terms—Clock synchronization, delay attack, IEEE 1588, physical-layer security, Schumann resonance, sequential detection, time transfer.

I. INTRODUCTION

PACKET-BASED time transfer under IEEE 1588 (PTP) secures message *integrity* but not message *timing*. Section II shows the vulnerability is algebraic, not implementational: the two-way exchange cannot separate path asymmetry from clock offset, so a delay attack forges nothing and breaks no key. Four known remedies exist, being a second disjoint path, a holdover clock, an engineered-reciprocal medium, and out-of-band radio time. Each assumes a resource some nodes lack; Section II-C states the condition each one requires and the theorem that condition traces to. The unserved node is single-homed, carries a commodity oscillator, and sits where GNSS is denied, jammed, or indoors.

The architecture analyzed here needs four properties of its reference signal, none specific to the Schumann band:

R1, global availability without infrastructure; **R2**, bandwidth sufficient for the target precision; **R3**, spatial coherence at deployment baselines; and **R4**, infeasibility of coherent forgery at scale. Mains phase fails R1 across interconnects; geomagnetic pulsations fail R2; spheric time-of-arrival methods require per-event association. The Schumann band survives all four filters, and Section II-E reads the alternatives against the same list.

Each node timestamps acquisition with its own clock and transmits $M_i = \text{Auth}_K(H[x_i], t_i^{\text{acq}}, n_i)$, where H is a collision-resistant hash and n_i a sequence number. The trusted boundary is acquisition: everything formed after M_i , including queueing, routing, and deliberate delay, leaves M_i unchanged. Altering the record, its timestamp, or its ordering is a forgery in the cryptographic sense, and replay fails the sequence check. The delay attack therefore holds an object it cannot edit.

A. Contributions

- 1) An identifiability analysis locating the delay attack in the null space of two-way time transfer (Section II-A).
- 2) Quantification of achievable offset precision in two observation domains of the shared signal, with a measured parameter-domain floor (Section IV-A) and an information-theoretic waveform bound (Section IV-B).
- 3) A sequential-detection result, with proof, showing accumulated bias before detection scales as the cube root of the attack rate, and with its flicker-noise boundary disclosed (Section III-D; calibrated in Section IV-C).
- 4) Measured cross-continental coherence at 3002 km from public data, a pre-declared null battery (Section IV-D), and a frozen, hash-deposited protocol whose quantitative prediction was confirmed at fifty times, inside a preregistered window of thirty to one hundred (Section IV-F).
- 5) A threat-model decomposition in which cryptography answers forgery and physics answers delay, with neither alone covering both (Section III-A).

II. LITERATURE REVIEW

A. The Exchange is Rank Deficient

The PTP exchange records (t_1, t_2, t_3, t_4) and estimates

$$\hat{\theta} = \frac{1}{2} [(t_2 - t_1) - (t_4 - t_3)] = \theta + \frac{1}{2} (d_{MS} - d_{SM}). \quad (1)$$

One equation, two unknowns. Only the *difference* of the path delays is observable, and the symmetric-path assumption $d_{MS} = d_{SM}$ is what converts the measurement into an offset. An on-path adversary inserting δ into one direction shifts $\hat{\theta}$

The author is an independent researcher (e-mail: k2luecke@gmail.com).

Analysis code, preregistration text, and run artifacts are on public deposit; see Appendix A.

by $\delta/2$. Nothing in the packet changes; MACs and signatures verify; the attack lives entirely in the unprotected physical observable of arrival time.

Equation (1) organizes the review that follows. Every defense surveyed below attempts to recover the missing rank from inside the exchange, and each carries a precondition that follows from what it must observe in order to do so.

B. The Vulnerability is Demonstrated, Not Hypothetical

Barreto *et al.* built a hardware delay box that advances or retards a slave clock by a few microseconds, and reported it undetectable both from the protocol's perspective and by the slave's clock-servo algorithm, independent of the cryptographic protection in use [3]. Finkenzeller *et al.* subsequently demonstrated static and incremental variants against PTP at microsecond precision on a testbed [7].

Encryption does not close the gap. Annessi *et al.* showed that statistical traffic analysis identifies PTP messages inside IPsec-, MACsec-, or TLS-protected traffic well enough to support selective delay, and identified the underlying conflict: the deterministic traffic pattern that makes a synchronization protocol precise is the same pattern that makes its packets identifiable for selective delay [4]. Obfuscation trades precision for concealment in both directions.

C. Packet-Domain Defenses and Their Preconditions

Mizrahi's game-theoretic treatment makes the boundary of the existing design space explicit [1]. An attacker chooses insertion delays (δ_{MS}, δ_{SM}); a defending process decides, at the end of each exchange, whether to accept or discard it. The defender's strategy set is $S_B = \{f \mid f : \mathbb{R}^4 \rightarrow \{\text{Pass}, \text{Drop}\}\}$, the space of functions of the four timestamps and nothing else. That definition, rather than any particular f , is what the present architecture contests. Round-trip thresholding, multiple clock sources, and path redundancy are three choices of f ; all three read arrival times, and (1) is the statement that arrival times do not contain the missing rank.

Each choice of f carries a sharp precondition, summarized in Table I.

The threshold condition is Mizrahi's: a drop threshold D_{RT} admits an equilibrium only when $2\varepsilon_{\max} \leq D_{RT} \leq \Delta_{ERR} + 2\varepsilon_{\min}$ for delays confined to $[\varepsilon_{\min}, \varepsilon_{\max}]$. That is, only when the accuracy requirement exceeds twice the delay spread, which is satisfiable in a low-variance single-hop setting such as an 802.1AS bridged network but not in the wide-area case. Path redundancy inherits $N > 3m$ from Lamport and Melliar-Smith [2], and the accompanying bound on residual error of the surviving paths is, by its author's own statement, not strictly a formal claim but rather an intuition.

PTP is weaker here than NTP. Round-trip detection depends on time distribution and delay measurement being coupled within one exchange, as they are in NTP; PTP does not guarantee that coupling. Under the peer-to-peer variant the Pdelay handshake is independent of the Sync messages that carry time, so an adversary who delays only Sync leaves the measured path delay untouched. Under end-to-end, the standard permits Delay_Req at a lower rate than Sync, leaving

Sync messages applied to the clock with no corresponding measurement, admitting the same evasion. The recommended countermeasure, end-to-end measurement at the full Sync rate with the clock update deferred to Delay_Resp, restores the coupling but not the rank: it makes a threshold meaningful in those cases where a valid threshold exists at all.

D. State of the Art

The state of the art has not moved off these preconditions. PTPsec [8] is the most complete treatment. It entangles detection with synchronization by invoking a measurement packet on every PTP event message, derives cyclic round-trip asymmetry over redundant paths, integrates into IEEE 1588-2019 [10], and detects both static and incremental attacks on hardware within a few synchronization intervals. Its own stated condition is that redundant paths be available in the network, and the authors note this is already enforced by standards such as TSN, which is precisely the deployment class the unserved node of Section I is not in. The 2025 journal extension reinforces the point rather than relaxing it [9]: it adds software-defined networking for dynamic topology discovery, redundant-path management, and deterministic routing, machinery whose necessity is itself a measure of what the redundancy precondition costs to satisfy. The 2012 conditions therefore still bound the current art; what has changed is the quality of the defense available to nodes that can already meet them.

Patching PTP cannot repair rank deficiency. That the missing information must come from outside the protocol is not a new claim: Barreto *et al.* stated it as the conclusion of the delay-box work [3], and Annessi *et al.* reached it independently [4], adding that knowledge of one-way delays and of the physical constraints of the network is what would permit a guaranteed bound on slave offset. This paper contributes not the diagnosis but a specific exterior: a reference that is globally available, requires no infrastructure, is not carried by the network, and admits a measured rather than assumed bound (Section IV). In the terms above, the claim is an enlargement of S_B , not a better element of it.

E. Alternative Reference Signals

Read against R1–R4, the candidate exteriors sort as follows. Mains phase is regional and fails R1 across interconnects, demonstrated incidentally by the grid-based null in Section IV-D. Geomagnetic pulsations and the millihertz geomagnetic art [21] lack bandwidth for sub-millisecond work and fail R2. Spheric time-difference-of-arrival methods [22] require per-event association and share the source-geometry systematic without the continuous integrability of the resonant band. GNSS is the out-of-band remedy whose denial defines the unserved node.

The Schumann band [18], [19] is not argued to be unique. It is the candidate that currently survives all four filters, and the architecture accepts any reference that does.

TABLE I
PACKET-DOMAIN DEFENSES AND THE PRECONDITION EACH REQUIRES

Class	Mechanism	Precondition	Where it fails
Threshold	compare d_{RT} to a bound	$\Delta_{ERR} \geq 2(\varepsilon_{\max} - \varepsilon_{\min})$, and d_{RT} must move under attack	wide-area delay spread; d_{RT} need not change at all
Guard / report	monitor node, or timestamped reports to a reference	report traffic indistinguishable from protocol traffic	attacker distinguishes custom event messages and treats them differently
Path redundancy	compare asymmetry across N paths	$N > 3m$ <i>verified</i> -disjoint paths	single-homed node; disjointness unverifiable (Sec. V-B)
Shared reference (<i>this work</i>)	correlate a global field carried in message content	in-band sensor meeting R1–R4 (Sec. I)	measured envelope (Sec. IV, V-C)

TABLE II
ERROR BUDGET FOR (3)

Term	Per event	After 1 h	Character
Bulk geometry	2.2 ms	subtracted	deterministic
Source location (50 km)	220 μ s	5 μ s	statistical
Dispersion residual	$\leq 163 \mu$ s	$\propto N^{-1/2}$	statistical
Day–night asymmetry	—	not averaged	systematic
Timestamp / DAQ	inst. dep.	inst. dep.	systematic

Per-event geometry and dispersion figures are per 500 km of effective path difference. Averaging assumes about 1800 events per hour.

III. METHODOLOGY

A. Architecture: a Fiducial in Content, Not Arrival

1) *Observation model*: Both nodes observe filtered, delayed, noisy copies of one realization of the cavity field $s(t)$:

$$x_A(t) = (h_A * s)(t - \theta_A) + n_A(t), \quad (2)$$

and likewise at B . The realization, not the mode parameters, is the fiducial; the parameters only set how much of it is usable.

2) *Exchange*: Records are exchanged as authenticated objects M_i . A network adversary can delay the arrival of M_A arbitrarily; the waveform inside, and its bound acquisition timestamp, do not move. Cross-correlating the records recovers

$$\hat{\tau}_{AB} = \theta_{AB} + \frac{d(S, A) - d(S, B)}{v(f)}, \quad (3)$$

where S is the effective source region and $v(f)$ the mode group velocity, 0.742 to 0.800 of c across the band [17]. The propagation term is geometry: roughly 2.2 ms per 500 km of effective path difference in bulk, with dispersion contributing up to 163 μ s per 500 km before single-parameter dedispersion.

3) *Error budget*: Statistical terms divide down with event count; systematic terms do not. The day–night asymmetry survives averaging and must be carried explicitly (Table II).

4) *Coherent forgery at scale*: Reproducing the cavity realization at one victim sensor requires a local field matching the global one in band. Doing so *consistently across geographically separated verifiers* requires either planetary-scale source power injected into a cavity of quality factor $Q \approx 4$, or physical capture of every sensor, at which point the adversary has capabilities against which no timing protocol is a defense. Forgery of the *record* is the MAC’s job; forgery of the *global field* is priced by physics.

5) *Local injection*: The preceding argument prices global forgery and does not dispose of a local one. An adversary with physical proximity to a single sensor can receive the in-band field and re-radiate a delayed copy into that sensor’s near field. The natural field is 0.1 to 1 pT in band, so the power required is not exotic; the requirement is proximity, not scale.

Such an injection is not merely undetected. Adding δ at the sensor of node A adds δ to $\hat{\tau}_{AB}$, which is indistinguishable from θ_{AB} having shifted by δ . It is precisely the signature the verifier is built to read as a clock offset, so the monitor would issue a correction in the adversary’s favour. Nor does adding nodes help: an injection at one node perturbs every pair through that node identically and therefore cancels in any loop sum, so a three-node closure test is blind to it by the same rank deficiency that (1) describes. A causal repeater can only delay and never advance, but the constraint binds per node rather than globally: (3) is antisymmetric in A and B , so an adversary who can choose which site to reach can produce either sign.

Four discriminants remain, and they are partial. *Amplitude*, since a re-radiator must dominate the natural field locally rather than merely add to it, and a partial-amplitude injection sums with the real field and appears as structure in the cross-spectrum. *Near-field polarization geometry*, since a local loop presents a field structure that a multi-axis sensor sees differently from the cavity’s horizontal-mode structure, which is the same physics as the vertical-coil null of Section IV-D. *Ramp detection*, since a growing injection is what Section III-D is for. And *sustained physical access*.

A static, well-matched injection installed at commissioning defeats all four. The honest statement of what the architecture achieves is therefore a relocation rather than an elimination: a remote on-path network adversary is converted into an on-site radio-frequency adversary, with a correspondingly different and considerably more expensive access requirement. That is a real security improvement and it is not the same claim as unforgeability.

6) *Division of labour*: Forge the global field, and source-power physics prices it. Forge the record or its timestamp, and authentication catches it. Replay, and sequence numbers catch it. Delay in transport, and correlation reads content rather than arrival. Creep below a threshold, and Section III-D bounds the gain. Inject locally, and the residual above stands. Cryptography and physics each cover the column the other cannot.

B. Parameter-Domain Estimation

Mode parameters, being frequencies, amplitudes, and quality factors, are fitted to a four-year Sierra Nevada record of 201534 ten-minute records [20], and parameter agreement is used as a cross-node check. The archival cadence is ten minutes, and the measured spectrum implies a process correlation scale of about fourteen minutes; both bear on the floor reported in Section IV-A.

C. Waveform-Domain Bound

Offset recovery is treated as passive time-delay estimation over the whitened 6 to 25 Hz band. The Weiss–Weinstein and Quazi small-error bound applies, with effective bandwidth $\beta \approx 13$ Hz inside $B = 19$ Hz. The general bound evaluated throughout is

$$\sigma_\tau(T) = \left[2T \int_6^{25} (2\pi f)^2 \frac{\gamma^2(f)}{1 - \gamma^2(f)} df \right]^{-1/2}, \quad (4)$$

evaluated with the measured, bias-corrected $\gamma^2(f)$.

Where Section IV evaluates (4) with *canonical* coherence, the quantity should be read as a Carter-style information bound [11] on the dominant coherent mode of the two-vector system. It is a defensible lower bound on the total available timing information, since canonical γ^2 is the coherence of one optimized linear combination. The exact multi-input multi-output generalization is not derived here; see Section V-D, route 8.

D. Sequential Detection

A static threshold invites the parking attack: ramp to just beneath the alarm line and hold there indefinitely. Rate-dependent first-crossing values run from 171 to 420 ns across the tested range. Sequential detection on the offset innovation changes the economics.

Proposition 1. *Let the adversary ramp the offset at constant rate r against an innovation sequence of scale σ sampled at cadence Δt , and let the detector alarm when the accumulated signal-to-noise ratio of a ramp hypothesis reaches κ . Then the bias accumulated before detection is*

$$\theta_{\text{det}} = \left(\kappa \sigma \sqrt{3\Delta t} \right)^{2/3} r^{1/3}. \quad (5)$$

Proof: Sample the ramp at $t_i = i \Delta t$ for $i = 1, \dots, N$, so the deterministic component is $s_i = r i \Delta t$. The matched-filter energy is

$$E = \sum_{i=1}^N s_i^2 = r^2 \Delta t^2 \frac{N(N+1)(2N+1)}{6} \rightarrow \frac{r^2 \Delta t^2 N^3}{3}$$

for large N . Against white innovations of variance σ^2 , the accumulated signal-to-noise ratio is E/σ^2 , and the alarm condition $E/\sigma^2 = \kappa^2$ gives $N^3 = 3\kappa^2 \sigma^2 / (r^2 \Delta t^2)$. Writing $T = N \Delta t$,

$$T = \frac{(3\kappa^2 \sigma^2 \Delta t)^{1/3}}{r^{2/3}}, \quad \theta_{\text{det}} = rT = (3\kappa^2 \sigma^2 \Delta t)^{1/3} r^{1/3},$$

which is (5). ■

The bias accumulated before detection therefore *falls* as the attack slows: halving r buys only a 21 % smaller pre-detection bias. Calibration is in Section IV-C.

Remark 1 (Disclosed boundary). *The proof assumes white innovations. Under flicker-type innovations, in the measured spectral exponent regime of 0.107, the advantage is bounded rather than eliminated: accumulated signal-to-noise grows more slowly than the $T^{3/2}$ implied by $E \propto N^3$, and the cube-root law softens. Route 5 of Section V-D treats detector design under $1/f$ as an open problem. Note also that κ is a threshold in signal-to-noise units, so at fixed false-alarm rate it depends on the effective number of independent samples and hence on the innovation autocorrelation; Section V-A takes up the consequences.*

E. Instrument

A single-axis induction coil with a low-noise preamplifier, sampled at a few hundred hertz against a GPS-disciplined or holdover-disciplined ADC clock, suffices to record the band. The vault magnetometers used below represent the low-noise end of the same commodity class. Sensor self-noise enters (4) only through $\gamma^2(f)$.

F. Data

The Gravitational Wave Open Science Center (GWOSC) auxiliary-channel releases [15], [16] include magnetometers at the LIGO Hanford (H1) and Livingston (L1) observatories, 3002 km apart, with GPS-disciplined timestamps. They constitute an accidental two-node testbed. Schumann-band content in these channels is independently documented in the detector-characterization literature [13], [14].

The *pilot* uses the GW170814 three-hour auxiliary release: 2.63 h of coincident data in two disjoint epochs on 2017-08-14. Livingston provides a low-noise vault magnetometer of the class deployed for Schumann monitoring; Hanford, in that release, provides only facility fluxgates whose self-noise exceeds the cavity signal several-fold. The pairing is a worst-case sensor test, and every pilot coherence reported below is a lower bound on what matched sensors achieve.

The *follow-on* uses the O3 auxiliary release, which carries vault magnetometers at *both* sites, at 4096 Hz via NDS2.

G. Estimators and Bias Handling

Magnitude-squared coherence is computed by Welch's method with 16 s Hann segments at 50 % overlap. Two bias references are reported side by side: the analytic $1/N$ floor, and an empirical null from pre-declared off-band windows. Detection is scored as excess over the empirical null in units of its scatter.

Sensor axes at the two sites are not aligned, so scalar pair coherence understates field coherence. The primary statistic is therefore the canonical coherence

$$\gamma_{\text{can}}^2(f) = \lambda_{\text{max}}[S_{xx}^{-1} S_{xy} S_{yy}^{-1} S_{yx}] \quad (6)$$

over the two-axis records: rotation-invariant, standard in magnetotellurics, and fixed a priori, which distinguishes it from

selecting whichever scalar pair produces the largest peak. The full 2×2 scalar-pair matrix and its maximum are reported as secondary.

Canonical coherence carries a larger finite-sample bias than single-pair magnitude-squared coherence. *Correction is therefore subtraction of the measured empirical off-band null mean*, not an analytic floor, and every bound is reported for both raw and corrected values. The verdict category is required to agree between them; disagreement is declared SENSITIVE. Every corrected coherence quoted in Section IV is the raw value less the off-band null mean stated alongside it. Null and surrogate construction, and the reasoning behind each choice, are in Appendix A-B.

H. Preregistration

The O3 analysis was frozen before any O3 sample was seen. The frozen text declares five gates, their thresholds, the surrogate constructions, and a pre-declared reading for every pass-fail branch including branches that did not occur. It was frozen as executable code rather than as a description of intent, so declared thresholds cannot drift after data are seen without changing the deposited hash, and the executing build records that hash in its output. Three rounds of external adversarial review preceded the freeze. Thresholds appear in the first column of Table III; governance rationale, construction detail, branch readings, and the full artifact chain are in Appendix A.

IV. RESULTS

A. Parameter-Domain Floor

Using parameter agreement as a cross-node check yields an offset floor of 90 s. The floor is set by the process correlation scale of about fourteen minutes implied by the measured spectrum, and is not improved by longer integration.

[Author decision: choose one mechanism and verify against residuals. (a) Cadence-limited: the floor is consistent with the interpolation limit of the ten-minute archival cadence, since about 0.15 of the 600 s interval reproduces 90 s; under this reading a faster-cadence record lowers it. (b) Systematic-limited: the residual is dominated by a station-dependent diurnal systematic, with the fourteen-minute scale setting where averaging stops helping and the diurnal term setting where it stops mattering.]

The mode statistics carry an internal consistency check. The three modes are mutually uncorrelated at short timescales, with $r_{13} = 0.003$, and propagating the measured mode variances and covariances predicts an f_3/f_1 ratio scatter of 2.60 % against 2.66 % observed, reproducing f_3/f_2 to the reported precision. The parameter domain is thus a coarse but *self-consistent* integrity channel, which is why the waveform domain is worth five to six orders of magnitude more.

B. Waveform-Domain Envelope

At unity cross-node coherence, (4) gives $57 \mu\text{s}$ at one hour and $12 \mu\text{s}$ at one day, degrading to $363 \mu\text{s}$ at one day for SNR-limited reception. These are *reference* numbers conditioned on assumed coherence. The remainder of this section replaces the assumption with measurement.

C. Detection Bias

Calibrated on the four-year record, $\kappa \approx 3.7$ predicts 20 and 202 ns for the two tested cadences, against 27 and 240 ns measured.

[Author decision: state the provenance of the measured values. If 27 and 240 ns come from the same four-year record on which κ was fitted, this is an internal consistency check and must be labelled as one; a labelled consistency check survives review, one presented as validation does not. If out-of-sample, say so, since it is materially stronger than it currently reads. State the value of σ used, and give both residuals as percentages. Section V-A depends on this answer.]

D. Pilot: Detection and Null Battery

The best single pair detects the fundamental at $\gamma^2(f_1) = 0.0091$, at $z = 7.8$ above the off-band null; stacking stations raises this to $z = 11$. The combined record detects all three modes, mode-ordered at $z = 19.8, 4.8$, and 3.5 , the spatial-response ordering the characterization of Section II-C predicts.

Four pre-declared scored nulls passed. Phase randomization collapses the fundamental to 0.0020. The vertical coil at Livingston, which horizontal-mode field structure predicts should not participate, is suppressed twentyfold. A fluxgate-by-fluxgate cross-site pair shows no excess. The disjoint second epoch independently replicates at $z = 14.2$. Mains contamination is excluded by geography: the sites sit on the Western and Eastern Interconnections, which are not phase-locked.

A fifth check is reported and *not scored*. A relative shift of 60 s collapses the fundamental to 0.0015, and the shift is valid as a collapse test for this estimator because it far exceeds the analysis segment and therefore pairs disjoint windows of any stationary process. It is not scored because population-level magnitude-squared coherence is shift-invariant, so the argument is avoidable and is not relied upon (Appendix A-B).

Applied to the pilot with identical parameters, the corner-station pairing gives $\gamma_{\text{can}}^2(f_1) = 0.0103$ raw against an empirical null of 0.0045 ± 0.0006 , hence **corrected 0.0058**, at $z = 9.7$, with all nine station-by-epoch cells independently significant at $z = 3.3$ to 16.5 . This is the denominator of the preregistered O3 prediction.

E. Pilot: Bound with Measured Coherence

Substituting the measured, bias-corrected pilot coherence into (4) gives $903 \mu\text{s}$ at one hour, measured rather than assumed, and under stationarity projects to $184 \mu\text{s}$ at one day and $34 \mu\text{s}$ at thirty days, through the worst-case sensor. What these figures should be compared against is treated in Section V-A.

F. O3: Preregistered Outcome

Results are summarized in Table III. The off-band null is 0.0399 ± 0.0118 across 2699 Welch averages, which produces both the corrected value and z per Section III-G.

TABLE III
PREREGISTERED GATES AND OUTCOMES. GPS 1264298000+6 H, 2020-01-29, 100 % RETRIEVAL

Gate	Frozen threshold	Outcome
G0 data	$\geq 90\%$ of requested samples per channel	PASS , 100 %
P1 coherence	$\gamma_{\text{can}}^2(f_1)$ exceeds off-band null at $z \geq 5$, exceeds all 20 block-permutation surrogates, and collapses under phase randomization	PASS . $z = 24.5$; observed 0.3291 against surrogate maximum 0.0065; phase-randomized 0.0014
P2 sensor	raw $\gamma_{\text{can}}^2(f_1) \geq 0.10$. Scored secondary: R , corrected value over the like-for-like pilot baseline 0.0058, forecast 30 to 100	UPHELD . Raw 0.3291, corrected 0.2892, R = 50 , inside the forecast window
D1 bound	sub-millisecond at measured T , reported for raw and corrected coherence with projections labelled	PASS . 45.7 μs raw and 76.6 μs corrected at $T = 6$ h; 22.8/38.3 μs at one day
D2b lag	majority of 30 min windows at empirical $p \leq 0.05$ against 19 same-statistic surrogates; median lag within 15 ms of the propagation band; robust between-window scatter $\leq \max(3 \times \text{per-window bound}, 0.3 \text{ ms}) = 0.795 \text{ ms}$	not passed . 11 of 12 windows at $p \leq 0.05$; median lag -3.85 ms ; scatter 4.95 ms

1) *Why the canonical statistic earned its place:* The pair-coherence matrix shows the cross-axis pairs dominating: H1Y $\times$ L1X at 0.158 and H1X $\times$ L1Y at 0.148, against diagonal values of 0.022 and 0.097. The sensor axes are near-orthogonal between sites, and the rotation-invariant estimator (6) recovers 0.329 where the best scalar pair sees 0.158. All three modes are again mode-ordered, at 0.329, 0.108, and 0.020.

2) *The lag is (3), measured:* Both records are GPS-disciplined, so $\theta_{AB} \approx 0$ and the recovered lag is, to the 0.265 ms per-window estimation precision, the propagation term of (3) itself. The median of -3.85 ms and the window range of $\pm 10 \text{ ms}$ sit inside the 13 ms envelope the 3002 km baseline permits at $v \approx 0.77c$ (Fig. 1b). Attribution to specific source longitudes is left to route 1 of Section V-D.

3) *Reading the D2b line:* The verdict stands as written. The frozen scatter rule assumed static source geometry over six hours, and a robust scatter (scaled median absolute deviation) of 4.95 ms exceeds 0.795 ms. What failed is informative: eleven of twelve windows individually beat all nineteen of their surrogates, and the per-window estimation bound is 0.265 ms, so the between-window scatter is 18.7 times estimation noise.

Exploratory, and labelled as such: $\gamma^2(f_1)$ trends upward across the morning (Fig. 1a), and the lag track stays inside the geometric band while migrating. Both are the signature of a moving effective source region, the (3) term acting on six-hour timescales, and they constitute the empirical justification for the systematic row of Table II. The pre-named chimney-direction comparison converts this from residual into observable.

4) *Provenance:* The executing build records its own hash in every output, and that hash matches the public deposit. Spectral matrices are well conditioned over the scored band: in the pilot f_1 band the minimum eigenvalue is positive in every bin, with condition number median 1.5 and maximum 1.8. Full artifact chain, version history, execution record, and known defects in the frozen deposit are in Appendix A.

V. DISCUSSION AND CONCLUSION

A. What the Bounds Mean

PTP's nominal accuracy target is sub-microsecond, and read against it the 184 μs figure of Section IV-E looks unserious.

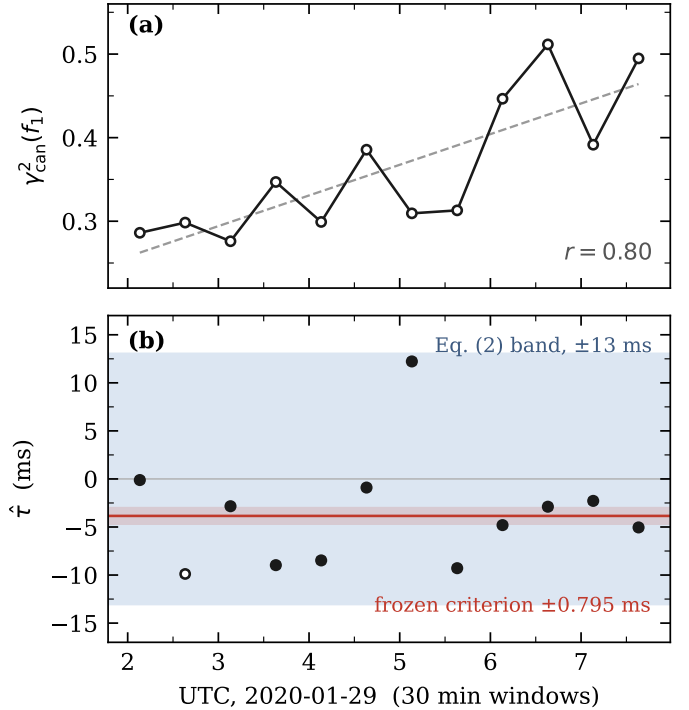


Fig. 1. Per-window structure of the O3 run, 12 consecutive 30 min windows. (a) Canonical coherence at the fundamental rises monotonically in trend across the morning ($r = 0.80$). (b) Recovered lag. Filled markers are windows exceeding all 19 surrogates ($p \leq 0.05$); the open marker is the one that does not. Error bars are the 0.265 ms per-window estimation bound and are smaller than the marker. The shaded blue region is the $\pm 13 \text{ ms}$ envelope permitted by (3) at the 3002 km baseline; the red line and narrow band are the median and the frozen $\pm 0.795 \text{ ms}$ scatter criterion. The lag stays inside the geometric envelope while migrating across it: the wander is 18.7 times estimation noise, and is signal.

But the nominal target is not the adversarial one, and for the node this architecture addresses the adversarial target does not exist.

Section II-C established that every element of the packet-domain defender's strategy set fails its precondition at a single-homed node on a wide-area path. No valid round-trip threshold exists at that delay spread. Redundancy requires paths the node does not have. Report traffic is distinguishable by an attacker watching for it. And both PTP variants permit an

evasion leaving the measured path delay untouched. Annessi *et al.* arrive at the same place from the other direction [4], concluding that a guaranteed bound on slave offset would require knowledge of one-way delays and of the physical constraints of the network: no such bound is derivable from the observables that node possesses.

The figures of Section IV are therefore not a competitive accuracy claim. They are the first measured bound of any size for a node whose guaranteed-under-attack precision was previously not a large number but an unobtainable one. Which applications a given σ_τ serves is a deployment question and is out of scope here.

1) *The two precision quantities:* A reader holding Sections IV-E and IV-C has two numbers differing by three orders of magnitude, and their relation must be stated rather than inferred.

σ_τ is the geophysical monitor's own offset-estimation precision at integration time T : the resolution with which an independent verification layer bounds a node's offset, and therefore the smallest static or step displacement it can certify against. θ_{det} is the bias a ramping adversary accumulates before the sequential detector alarms on the *servo's* offset-innovation sequence. The constant κ is calibrated on the four-year record, but the σ in (5) is the servo innovation scale, which is why θ_{det} is reported in nanoseconds and σ_τ in microseconds. The monitor sits beneath the servo rather than in place of it: the servo supplies precision, and the monitor bounds how far that precision can be displaced without the displacement becoming observable.

2) *Transfer of κ :* Two limits are stated here rather than deferred. By Proposition 1, κ is an alarm threshold in signal-to-noise units, so at fixed false-alarm rate it depends on the autocorrelation of the innovation sequence and not on its scale alone. The calibrating record has a measured spectral exponent of 0.107, so κ is fixed under mildly correlated innovations and applied to a sequence treated as white. Because positive correlation reduces the effective independent-sample count, this transfer inflates κ relative to its white-innovation value and renders θ_{det} conservative in that respect.

[Author decision, contingent on Section IV-C. Cutting the other way, the predicted values fall below the measured at both tested cadences, and underprediction of achievable pre-alarm bias is the unsafe direction for a security quantity. If Section IV-C resolves out-of-sample, state that κ is accurate to within tens of percent with the residual in the direction that understates achievable bias, and that out-of-sample validation against an independent servo innovation trace remains outstanding. If it resolves in-sample, this paragraph cannot claim a validated residual and must state that no out-of-sample validation of κ has been performed.]

B. Independence: Physical Against Topological

R1–R4 govern *reference signals*. The nearest alternative architecture is path redundancy, which is not a reference at all, and the comparison turns on what its independence rests on.

Both approaches introduce a second information source against which the primary channel is checked. Path redundancy's independence is topological and assumed: it requires the adversary to occupy at most m of N paths, and disjointness at the logical layer does not imply disjointness in physical plant. Shared conduit, a shared transit autonomous system, or route manipulation can defeat it, and the slave generally cannot verify that they have not. The independence claimed here is physical: the cavity field is not carried by the network under any topology, so no arrangement of adversary position or route control brings it within reach.

Provisioning costs differ accordingly, being more than three verified-disjoint long-haul paths against one sensor per node, and so do the failure modes. Redundancy degrades silently when its disjointness assumption is false, whereas loss of cross-node coherence is observable at the node.

C. Limitations

1) *Measurement campaign:* The measured coherence comes from one baseline, three epochs totalling under nine hours, and two sensor classes. Both nodes are instruments of a single observatory network with homogeneous data-acquisition chains, so a commodity node of independent construction remains the outstanding independence test. Multi-baseline and multi-season coherence maps remain unmeasured. The one-day and thirty-day figures are $\sqrt{T}$ projections from measured six-hour data and are labelled as such; a 20 h coincident segment permits a direct check. Extracting a *stable* offset at the D1 level requires the source-geometry model, and the O3 run measured exactly the wander that model must remove. The sequential-detection guarantee weakens under flicker innovations (Remark 1). The 90 s floor awaits its mechanism decision. None of these is hidden inside an assumption; each is a named, testable route in Section V-D.

2) *Sensor and deployment envelope:* The natural field is 0.1 to 1 pT in band, and the limiting factor in deployment is rarely the intrinsic noise floor of a modern induction coil. It is the local environment: mains harmonics and switching electronics, microphonics from wind, rain, or footfall, vibrating metal, and proximity to power infrastructure. The two configurations measured here bracket this envelope rather than sampling its best case. Vault-sited coils resolved corrected $\gamma^2(f_1) = 0.29$. Facility fluxgates inside an operating interferometer building, an electromagnetic environment few industrial nodes will exceed for hostility, still carried a detectable channel at corrected $\gamma^2 = 0.0058$, at a measured precision cost of roughly five times at one day.

Site quality therefore buys precision; it does not gate existence, down to at least the facility-noise level measured. Below that level burial is real: the architecture is functional today where Schumann observatories already operate and in engineered quiet enclosures, and it is not yet a drop-in attachment for an arbitrary urban node. The gap is engineering, not physics: reference-sensor differencing, adaptive cancellation of tracked interferers, mains-harmonic notching, mechanical isolation, and pre-deployment site characterization, which is the same noise-spectrum measurement the detector's calibration requires.

Where mitigation fails, the semantics fail safe. Sudden loss of cross-node coherence is itself an integrity event, consequentially indistinguishable from jamming, and flags the node as unverified rather than silently trusted. This also keeps the monitor off the always-drop equilibrium of Mizrahi's game [1], in which a detector attains maximum payoff by rejecting every exchange while leaving the node with no time at all: here the reject condition is a measured physical event carrying a null battery behind it, not a threshold on a statistic whose false-alarm rate is set by network delay variance. "Commodity sensor" is retained in its precise sense, that no exotic instrument is required, and explicitly not in the sense of any sensor at any site.

D. Future Work

- 1) **Chimney-direction test.** Regress the per-window τ and γ^2 of Fig. 1 against independent global lightning activity over multi-day records. Agreement converts the failed static rule into a second measured observable and calibrates the (3) subtraction; disagreement bounds the source model. This is the highest information-per-hour test available.
- 2) **Source-geometry subtraction to a stable offset.** Implement Table II end to end: dedispersion, source-region estimation from the lag and coherence track itself, ensemble averaging, and explicit carry of the day-night systematic. Target is reduction of the 4.95 ms wander toward the 0.265 ms estimation floor.
- 3) **Multi-baseline, multi-season coherence atlas.** Repeat the frozen methodology across dedicated ELF stations at 500 to 10000 km and across seasons. This is the first dataset against which R3 can be certified rather than sampled.
- 4) **Direct one-day verification.** A single 20 h run tests the $\sqrt{T}$ projection against measurement with no new assumptions.
- 5) **Sequential detection under flicker innovations.** Close Remark 1: characterize the innovation spectrum from long records, then design and benchmark detectors whose pre-detection bias degrades gracefully under $1/f$. This concerns detector *design*; whether a constant calibrated on one innovation spectrum transfers to another is a separate question raised in Section V-A.
- 6) **Embedded implementation and protocol integration.** Build the reference node, measure its γ^2 against a vault-class station under progressively degraded siting, and specify the integration point with IEEE 1588 security mechanisms [10]. This route also carries holdover fusion, which the architecture answers compositionally rather than competitively: free-running oscillator error grows with elapsed time T while σ_τ falls as $T^{-1/2}$, so the node's guaranteed bound at any T is the minimum of the two, and that minimum is finite across a range over which neither source alone is. Holdover carries the short interval, the monitor the long one, and they are not in competition because they bound different things (Section V-A). No crossover time is quoted: a disciplined

oscillator's holdover begins from residual drift rather than raw fractional offset, and the dominant term in a field enclosure is thermal, so the crossover is a part-and-site band spanning an order of magnitude in T rather than a number. Measuring it is part of this route.

- 7) **Preregistration as reporting standard.** The evidence architecture used here is domain-independent. Appendix A is the worked instance, defects included.
- 8) **Exact MIMO generalization of the coherence bound.** Section III-C evaluates (4) with canonical coherence and reads the result as an information bound on the dominant coherent mode. Deriving the exact multi-input multi-output generalization would establish how much timing information the subdominant modes carry, and therefore whether the canonical evaluation is conservative by a negligible or a material factor.

E. Conclusion

Two-way time transfer will remain rank deficient; the question is where the missing information comes from. This paper's answer is a globally excited, spatially correlated field whose realization rides inside authenticated records, out of reach of transport delay. The contribution is not a competitive accuracy figure but the existence of a measured bound for a node class whose guaranteed-under-attack precision was previously not derivable at all from the observables available to it.

Sequential detection removes the incremental attacker's advantage whenever offset innovations are white or short-range correlated: by Proposition 1 the bias achievable before detection scales as the cube root of the ramp rate and vanishes as the attack slows. Under flicker-type innovations the advantage is bounded rather than eliminated.

The coherence this architecture requires is no longer assumed. It is measured at 3002 km through the wrong sensor and through the right one, it survived its null battery twice, and the preregistered sensor-matching prediction was confirmed at fifty times, inside its window of thirty to one hundred, by an analysis whose hash was on public deposit before the data were seen. The separately preregistered static-lag-stability criterion was not met and is retained as a limitation rather than retroactively relaxed.

APPENDIX A

EVIDENCE GOVERNANCE AND PROVENANCE

Route 7 of Section V-D proposes formalizing this evidence architecture as a reporting standard for physical-layer security claims. This appendix is the worked instance. It is written to be checkable rather than persuasive, and it includes the defects.

A. What Was Frozen, and Why as Code

The O3 stage was preregistered as executable code rather than as a description of intent. The frozen script contains the protocol text, gates, thresholds, surrogate constructions, pre-declared readings, and a freeze clause, and it refuses to run if its own bytes do not hash to the deposited value.

Freezing analysis and protocol as one object removes the gap between what was declared and what was run. Three consequences follow: declared thresholds cannot drift after data are seen, because changing them changes the hash; every reported number carries the identity of the code that produced it; and a reader can verify the binding without trusting the authors, by hashing the deposited file.

B. Null and Surrogate Construction

Pre-freeze review changed four things. Each is recorded because each corrected a weaker design.

The primary statistic is canonical, not best-pair. Selecting whichever scalar pair produces the largest peak is pair selection under another name. Equation (6) is rotation-invariant and fixed a priori; the O3 pair matrix shows why this mattered.

N1 is a block permutation, not a time shift. For a stationary process a time shift multiplies the cross-spectrum by a unit-magnitude phase factor and leaves magnitude-squared coherence unchanged, so a shift null is not a valid collapse test for this estimator at the population level. The frozen N1 splits the L1 vector record into 64 s blocks and permutes block order, applying the same permutation to both L1 axes, preserving marginal spectra and within-site covariance while destroying inter-site temporal correspondence. Twenty surrogates give empirical $p < 1/21$.

The 60 s shift is reported and not scored, for the reason above. This is a property of the estimator, not of the dataset, so it applies to the pilot battery equally.

The D2b null uses the same four-pair statistic as the observation. An earlier design built the null from one shifted pair multiplied by four, a different statistic and therefore not a null for it. The frozen version passes 19 block-permutation surrogates through the identical four-pair construction per window.

C. Pre-Declared Branch Readings

Declared before execution, for branches that did and did not occur. P1 pass with P2 fail reads as “channel real, sensor- or κ^2 -limited, not model-false.” Disagreement between raw and corrected verdicts is declared SENSITIVE. Retrieval below 90 % is INCONCLUSIVE with no scoring. Scoring the lag against zero is prohibited: it is scored against the (3) propagation band, since GPS discipline removes clock offset but not propagation. The last is the branch most likely to be assumed; two GPS-disciplined stations have no reason to see zero lag, and testing against zero would have been the wrong hypothesis.

D. Analysis Versions

[Author decision. If the byte-exact v4.1 is recovered, Table IV stands and the third row is a separate logged change. If not, v4.1 was superseded before it ever ran to completion, the reconstruction is a new version with its own hash, and this table must say so. Presenting a freshly written patch under an already-published hash is the one move that would damage the provenance chain this appendix establishes.]

TABLE IV
VERSION IDENTITIES

Version	Change	Statistics affected
v4 f45dca61...	Frozen deposit; produced all reported O3 statistics	—
v4.1 98a27d1b...	Post-verdict serialization fix	none
[pending]	Serialization of the off-band null mean and standard deviation	none

E. Execution Record

Environment: Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1, nds2-client 0.16.12, retrieved via NDS2 from GWOSC. Data: GPS 1264298000, duration 21600 s, 100 % retrieval, four channels, analysis rate 128 Hz, Welch 2048-sample segments with 1024 overlap and 2699 averages.

An offline gate ran before the network fetch and was required to pass: recovery of a synthetic coherence of 0.30 within 15 %; recovery of a known lag of +31.250 ms within 0.5 ms; zero-lag centring within 0.5 ms; and an end-to-end test carrying a synthetic vector field through a 40° sensor rotation, additive noise, the four-pair statistic, the permutation null, and lag extraction. All passed.

Incomplete termination. The v4 run computed every statistic, printed the verdict block, and wrote its results file, then raised a `NameError` at the array-archive write on the final line of the scoring function. No reported statistic is affected, since the failure occurs after all are computed and serialized, and the traceback line number establishes this independently of any claim by the author. The spectral archive was not written by that run. *[Author decision: state whether the rerun produced it.]*

F. Known Defects in the Frozen Deposit

Recorded rather than corrected, because correcting them would break the freeze.

The D2b pass line is stated two ways. The protocol defines per-window detection as the observed maximum Q exceeding all 19 surrogates at $p \leq 0.05$, and the summary pass line reads “window SNR at least 5 in a majority of windows.” The scoring code implements the first. The ambiguity is immaterial: under the p -value rule 11 of 12 windows qualify, under the detection-statistic rule 8 of 12 qualify, both are majorities, and D2b failed on the between-window scatter criterion under either reading. It is disclosed because a reader of the deposit will find it, not because it changes anything.

The corrected coherence is not recoverable from the deposited results file. The off-band null mean is computed and printed but not serialized, so a reader given only the results file can obtain the raw value but not the corrected one, and cannot reproduce R without rerunning. Addressed by the pending version in Table IV.

G. Reproduction

Seeds are fixed and the input data are immutable public records, so a rerun of the same version should reproduce the

reported verdict exactly; any difference indicates environment drift and is itself diagnostic. A reproduction notebook accompanies the paper which verifies the freeze hash before running, archives prior outputs before overwriting them, runs the offline gate, and diffs recomputed results against recorded ones field by field, separating statistical differences from provenance differences.

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